3.Single Value Decomposition

November 4, 2026

[2]:
```
import os
import pandas as pd
```

[4]:
```
iris1 = pd.read_csv("iris.csv")  # Cargar la base de datos de IRIS
iris1
```

[4]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
..	…	…	…	…	…
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

[150 rows x 5 columns]

[5]:
```
etiquetas = iris1["variety"]
etiquetas
```

[5]:
```
0         Setosa
1         Setosa
2         Setosa
3         Setosa
4         Setosa
          …
145    Virginica
146    Virginica
147    Virginica
148    Virginica
149    Virginica
Name: variety, Length: 150, dtype: object
```

[6]:
```
from numpy import array
from sklearn.decomposition import TruncatedSVD
A = iris1
# Remover columna no numérica
A = A.drop(["variety"], axis = 1)
A
```

[6]:

	sepal.length	sepal.width	petal.length	petal.width
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
..	…	…	…	…
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

[150 rows x 4 columns]

[7]:
```
from numpy import diag
from numpy import zeros
from scipy.linalg import svd
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

[8]:
```
U, s, VT = svd(A)   # Obtención de matrices cuyo producto arroja la matriz A
Sigma = zeros((A.shape[0], A.shape[1]))   # Creación de matriz de ceros de la␣
 ↪misma dimensión que A
Sigma[:A.shape[1], :A.shape[1]] = diag(s)
```

[9]:
```
print("Matriz U:")
print(U)
```

```
Matriz U:
[[-0.06161685  0.12961144  0.0021386  … -0.09343429 -0.09573864
  -0.08085465]
 [-0.05807094  0.11101978  0.07067239 …  0.03690405 -0.03153954
   0.01309526]
 [-0.05676305  0.11796647  0.00434255 …  0.03066199  0.19531473
   0.13569909]
 …
 [-0.0940593  -0.0498297  -0.04144001 …  0.98181631 -0.02194514
  -0.00894446]
```

```
 [-0.09488961 -0.05610123 -0.21297821 … -0.02155617  0.94178018
  -0.02971961]
 [-0.08847836 -0.0515697  -0.09575285 … -0.0086052  -0.03021088
   0.9736599 ]]
```

[11]:
```
print("Matriz Sigma")
print(Sigma)
```

```
Matriz Sigma
[[95.95991387  0.          0.          0.        ]
 [ 0.         17.76103366  0.          0.        ]
 [ 0.          0.          3.46093093  0.        ]
 [ 0.          0.          0.          1.88482631]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        
```

```
[ 0.          0.          0.          0.        ]
[ 0.          0.          0.          0.        ]
[ 0.          0.          0.          0.        ]
```

```
[ 0.          0.          0.          0.        ]
[ 0.          0.          0.          0.        ]
[ 0.          0.          0.          0.        ]
```

```
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]
 [ 0.          0.          0.          0.        ]]
```

[12]:
```
print("Matriz VT")
print(VT)
```

```
Matriz VT
[[-0.75110816 -0.38008617 -0.51300886 -0.16790754]
 [ 0.2841749   0.5467445  -0.70866455 -0.34367081]
 [ 0.50215472 -0.67524332 -0.05916621 -0.53701625]
 [ 0.32081425 -0.31725607 -0.48074507  0.75187165]]
```

[13]:
```
# Para reconstruir la matriz original
B = U.dot(Sigma.dot(VT))
print(B)
```

```
[[5.1 3.5 1.4 0.2]
 [4.9 3.  1.4 0.2]
 [4.7 3.2 1.3 0.2]
 [4.6 3.1 1.5 0.2]
 [5.  3.6 1.4 0.2]
 [5.4 3.9 1.7 0.4]
 [4.6 3.4 1.4 0.3]
 [5.  3.4 1.5 0.2]
 [4.4 2.9 1.4 0.2]
 [4.9 3.1 1.5 0.1]
 [5.4 3.7 1.5 0.2]
 [4.8 3.4 1.6 0.2]
 [4.8 3.  1.4 0.1]
 [4.3 3.  1.1 0.1]
 [5.8 4.  1.2 0.2]
 [5.7 4.4 1.5 0.4]
 [5.4 3.9 1.3 0.4]
 [5.1 3.5 1.4 0.3]
 [5.7 3.8 1.7 0.3]
```

```
[5.1 3.8 1.5 0.3]
[5.4 3.4 1.7 0.2]
[5.1 3.7 1.5 0.4]
[4.6 3.6 1.  0.2]
[5.1 3.3 1.7 0.5]
[4.8 3.4 1.9 0.2]
[5.  3.  1.6 0.2]
[5.  3.4 1.6 0.4]
[5.2 3.5 1.5 0.2]
[5.2 3.4 1.4 0.2]
[4.7 3.2 1.6 0.2]
[4.8 3.1 1.6 0.2]
[5.4 3.4 1.5 0.4]
[5.2 4.1 1.5 0.1]
[5.5 4.2 1.4 0.2]
[4.9 3.1 1.5 0.2]
[5.  3.2 1.2 0.2]
[5.5 3.5 1.3 0.2]
[4.9 3.6 1.4 0.1]
[4.4 3.  1.3 0.2]
[5.1 3.4 1.5 0.2]
[5.  3.5 1.3 0.3]
[4.5 2.3 1.3 0.3]
[4.4 3.2 1.3 0.2]
[5.  3.5 1.6 0.6]
[5.1 3.8 1.9 0.4]
[4.8 3.  1.4 0.3]
[5.1 3.8 1.6 0.2]
[4.6 3.2 1.4 0.2]
[5.3 3.7 1.5 0.2]
[5.  3.3 1.4 0.2]
[7.  3.2 4.7 1.4]
[6.4 3.2 4.5 1.5]
[6.9 3.1 4.9 1.5]
[5.5 2.3 4.  1.3]
[6.5 2.8 4.6 1.5]
[5.7 2.8 4.5 1.3]
[6.3 3.3 4.7 1.6]
[4.9 2.4 3.3 1. ]
[6.6 2.9 4.6 1.3]
[5.2 2.7 3.9 1.4]
[5.  2.  3.5 1. ]
[5.9 3.  4.2 1.5]
[6.  2.2 4.  1. ]
[6.1 2.9 4.7 1.4]
[5.6 2.9 3.6 1.3]
[6.7 3.1 4.4 1.4]
[5.6 3.  4.5 1.5]
```

```
[5.8 2.7 4.1 1. ]
[6.2 2.2 4.5 1.5]
[5.6 2.5 3.9 1.1]
[5.9 3.2 4.8 1.8]
[6.1 2.8 4.  1.3]
[6.3 2.5 4.9 1.5]
[6.1 2.8 4.7 1.2]
[6.4 2.9 4.3 1.3]
[6.6 3.  4.4 1.4]
[6.8 2.8 4.8 1.4]
[6.7 3.  5.  1.7]
[6.  2.9 4.5 1.5]
[5.7 2.6 3.5 1. ]
[5.5 2.4 3.8 1.1]
[5.5 2.4 3.7 1. ]
[5.8 2.7 3.9 1.2]
[6.  2.7 5.1 1.6]
[5.4 3.  4.5 1.5]
[6.  3.4 4.5 1.6]
[6.7 3.1 4.7 1.5]
[6.3 2.3 4.4 1.3]
[5.6 3.  4.1 1.3]
[5.5 2.5 4.  1.3]
[5.5 2.6 4.4 1.2]
[6.1 3.  4.6 1.4]
[5.8 2.6 4.  1.2]
[5.  2.3 3.3 1. ]
[5.6 2.7 4.2 1.3]
[5.7 3.  4.2 1.2]
[5.7 2.9 4.2 1.3]
[6.2 2.9 4.3 1.3]
[5.1 2.5 3.  1.1]
[5.7 2.8 4.1 1.3]
[6.3 3.3 6.  2.5]
[5.8 2.7 5.1 1.9]
[7.1 3.  5.9 2.1]
[6.3 2.9 5.6 1.8]
[6.5 3.  5.8 2.2]
[7.6 3.  6.6 2.1]
[4.9 2.5 4.5 1.7]
[7.3 2.9 6.3 1.8]
[6.7 2.5 5.8 1.8]
[7.2 3.6 6.1 2.5]
[6.5 3.2 5.1 2. ]
[6.4 2.7 5.3 1.9]
[6.8 3.  5.5 2.1]
[5.7 2.5 5.  2. ]
[5.8 2.8 5.1 2.4]
```

```
 [6.4 3.2 5.3 2.3]
 [6.5 3.  5.5 1.8]
 [7.7 3.8 6.7 2.2]
 [7.7 2.6 6.9 2.3]
 [6.  2.2 5.  1.5]
 [6.9 3.2 5.7 2.3]
 [5.6 2.8 4.9 2. ]
 [7.7 2.8 6.7 2. ]
 [6.3 2.7 4.9 1.8]
 [6.7 3.3 5.7 2.1]
 [7.2 3.2 6.  1.8]
 [6.2 2.8 4.8 1.8]
 [6.1 3.  4.9 1.8]
 [6.4 2.8 5.6 2.1]
 [7.2 3.  5.8 1.6]
 [7.4 2.8 6.1 1.9]
 [7.9 3.8 6.4 2. ]
 [6.4 2.8 5.6 2.2]
 [6.3 2.8 5.1 1.5]
 [6.1 2.6 5.6 1.4]
 [7.7 3.  6.1 2.3]
 [6.3 3.4 5.6 2.4]
 [6.4 3.1 5.5 1.8]
 [6.  3.  4.8 1.8]
 [6.9 3.1 5.4 2.1]
 [6.7 3.1 5.6 2.4]
 [6.9 3.1 5.1 2.3]
 [5.8 2.7 5.1 1.9]
 [6.8 3.2 5.9 2.3]
 [6.7 3.3 5.7 2.5]
 [6.7 3.  5.2 2.3]
 [6.3 2.5 5.  1.9]
 [6.5 3.  5.2 2. ]
 [6.2 3.4 5.4 2.3]
 [5.9 3.  5.1 1.8]]
```

[14]:
```
# Reducción a DOS Dimensiones
n_elements = 2

UReduced2 = U[:, :n_elements]
print("U Reducida:")
print(UReduced2)
```

```
U Reducida:
[[-6.16168450e-02  1.29611444e-01]
 [-5.80709402e-02  1.11019776e-01]
 [-5.67630474e-02  1.17966465e-01]
 [-5.66534426e-02  1.05308145e-01]
```

```
[-6.12302023e-02  1.31089790e-01]
[-6.75031684e-02  1.30884835e-01]
[-5.74820774e-02  1.16598182e-01]
[-6.09726328e-02  1.20943120e-01]
[-5.37611959e-02  9.99414853e-02]
[-5.88266594e-02  1.12043088e-01]
[-6.52918230e-02  1.36578094e-01]
[-5.99417779e-02  1.13753142e-01]
[-5.71132323e-02  1.11354756e-01]
[-5.15957540e-02  1.15324791e-01]
[-6.80071908e-02  1.64183057e-01]
[-7.07625894e-02  1.59056485e-01]
[-6.53647385e-02  1.46844820e-01]
[-6.17918218e-02  1.27676473e-01]
[-6.92802967e-02  1.34541441e-01]
[-6.35146947e-02  1.32921488e-01]
[-6.51727724e-02  1.19363091e-01]
[-6.32935829e-02  1.27908181e-01]
[-5.59608478e-02  1.40649812e-01]
[-6.29534207e-02  1.05679869e-01]
[-6.15456003e-02  1.01783154e-01]
[-5.99228863e-02  1.04639775e-01]
[-6.18571938e-02  1.13083182e-01]
[-6.29341837e-02  1.27221439e-01]
[-6.20034877e-02  1.28133097e-01]
[-5.83668698e-02  1.05996477e-01]
[-5.87535125e-02  1.04518131e-01]
[-6.44535110e-02  1.23473141e-01]
[-6.51357378e-02  1.47626432e-01]
[-6.75203891e-02  1.57559767e-01]
[-5.90016362e-02  1.10108117e-01]
[-5.85766335e-02  1.26756434e-01]
[-6.42131623e-02  1.40001403e-01]
[-6.02724944e-02  1.31424770e-01]
[-5.36226769e-02  1.07009819e-01]
[-6.17553640e-02  1.22543111e-01]
[-6.04744831e-02  1.30066478e-01]
[-5.18077654e-02  8.51264784e-02]
[-5.44148539e-02  1.13166493e-01]
[-6.26032358e-02  1.12291578e-01]
[-6.58281013e-02  1.15026533e-01]
[-5.74631858e-02  1.07484815e-01]
[-6.38743254e-02  1.30866463e-01]
[-5.65149237e-02  1.12376478e-01]
[-6.45090918e-02  1.34978104e-01]
[-6.00419369e-02  1.21854779e-01]
[-9.50422391e-02 -4.11326398e-03]
[-8.94516138e-02 -7.66818721e-03]
```

```
[-9.51076111e-02 -1.87065550e-02]
[-7.58192461e-02 -2.59532167e-02]
[-8.91845985e-02 -2.23715412e-02]
[-8.20381882e-02 -2.73115299e-02]
[-9.03091628e-02 -1.61048038e-02]
[-6.72517650e-02  1.26006092e-03]
[-9.00134647e-02 -1.38232718e-02]
[-7.46957758e-02 -1.63848150e-02]
[-6.75193572e-02 -1.74332892e-02]
[-8.31419586e-02 -9.85482711e-03]
[-7.88118832e-02 -1.52266876e-02]
[-8.68093930e-02 -2.77481926e-02]
[-7.68400783e-02  1.00767816e-02]
[-9.06941347e-02 -2.15851353e-05]
[-8.23975875e-02 -2.66247878e-02]
[-7.97614708e-02 -7.02497949e-03]
[-8.39252666e-02 -4.16515405e-02]
[-7.65095933e-02 -1.03366139e-02]
[-8.76667106e-02 -3.34430418e-02]
[-8.24960756e-02 -9.61586108e-04]
[-8.80346931e-02 -4.67765227e-02]
[-8.60633510e-02 -2.69565882e-02]
[-8.68441800e-02 -5.05326495e-03]
[-8.95153150e-02 -4.69991309e-03]
[-9.24270302e-02 -2.36165904e-02]
[-9.40306212e-02 -3.28448113e-02]
[-8.51324237e-02 -2.33031619e-02]
[-7.53750063e-02  1.22366693e-02]
[-7.47961661e-02 -1.10249457e-02]
[-7.40865819e-02 -5.09997886e-03]
[-7.90422093e-02 -2.91492876e-03]
[-8.77228682e-02 -5.53347837e-02]
[-8.08321251e-02 -2.98247694e-02]
[-8.72878429e-02 -9.84644677e-03]
[-9.24729338e-02 -1.39265443e-02]
[-8.42195253e-02 -2.91132749e-02]
[-7.99092041e-02 -6.79486183e-03]
[-7.66114230e-02 -1.97965423e-02]
[-7.89709646e-02 -3.07432189e-02]
[-8.66708741e-02 -2.06798593e-02]
[-7.91807283e-02 -9.98326205e-03]
[-6.76384077e-02 -2.18285462e-04]
[-7.92555461e-02 -2.00198695e-02]
[-8.10515660e-02 -7.24989641e-03]
[-8.08304543e-02 -1.22632043e-02]
[-8.52787176e-02 -8.25324653e-03]
[-6.77844702e-02  1.75733973e-02]
[-7.98997583e-02 -1.13515454e-02]
```

```
[-9.88338505e-02 -8.53894903e-02]
[-8.66823361e-02 -6.43396776e-02]
[-1.02672920e-01 -7.00946964e-02]
[-9.38862295e-02 -6.81980592e-02]
[-9.76169023e-02 -7.76396157e-02]
[-1.10328828e-01 -9.00247153e-02]
[-7.52879803e-02 -5.70863507e-02]
[-1.05455794e-01 -8.01281242e-02]
[-9.65020152e-02 -8.20914370e-02]
[-1.07601304e-01 -6.57445578e-02]
[-9.43168735e-02 -3.96830270e-02]
[-9.24479381e-02 -6.27197251e-02]
[-9.81862967e-02 -5.89346842e-02]
[-8.47477972e-02 -7.00413173e-02]
[-8.79533083e-02 -7.09361942e-02]
[-9.51282875e-02 -5.50679223e-02]
[-9.53131729e-02 -5.79297443e-02]
[-1.14989851e-01 -6.97229940e-02]
[-1.11480981e-01 -1.16578003e-01]
[-8.50328416e-02 -6.48015027e-02]
[-1.01180373e-01 -6.30279529e-02]
[-8.46187241e-02 -5.84163005e-02]
[-1.10679013e-01 -9.66364242e-02]
[-8.93518003e-02 -4.64247606e-02]
[-9.96610459e-02 -5.92796557e-02]
[-1.04257505e-01 -6.05231151e-02]
[-8.84305502e-02 -4.09564181e-02]
[-8.89746034e-02 -4.03897307e-02]
[-9.47978024e-02 -7.54813178e-02]
[-1.02046160e-01 -5.48298557e-02]
[-1.04948198e-01 -7.55614491e-02]
[-1.14601538e-01 -5.06830826e-02]
[-9.49727792e-02 -7.74162886e-02]
[-9.02921735e-02 -4.55215034e-02]
[-9.04325947e-02 -7.28931692e-02]
[-1.08788476e-01 -7.23446854e-02]
[-9.69165324e-02 -6.44161979e-02]
[-9.49265302e-02 -5.64513979e-02]
[-8.76572648e-02 -3.79997254e-02]
[-9.88305089e-02 -5.02663601e-02]
[-9.88591917e-02 -6.72512462e-02]
[-9.75766400e-02 -4.21663133e-02]
[-8.66823361e-02 -6.43396776e-02]
[-1.01466857e-01 -7.26079359e-02]
[-1.00360953e-01 -6.70195388e-02]
[-9.61496966e-02 -5.24346281e-02]
[-8.92692075e-02 -5.85064019e-02]
[-9.40593040e-02 -4.98296974e-02]
```

```
 [-9.48896096e-02 -5.61012256e-02]
 [-8.84783560e-02 -5.15697045e-02]]
```

[15]:
```
SigmaReduced2 = Sigma[:n_elements,:n_elements]
print("Matriz Sigma Reducida")
print(SigmaReduced2)
```

```
Matriz Sigma Reducida
[[95.95991387  0.        ]
 [ 0.         17.76103366]]
```

[16]:
```
VTReduced2 = VT[:n_elements, :]
print("VT Reducida")
print(VTReduced2)
```

```
VT Reducida
[[-0.75110816 -0.38008617 -0.51300886 -0.16790754]
 [ 0.2841749   0.5467445  -0.70866455 -0.34367081]]
```

[17]:
```
AReduced2 = UReduced2.dot(SigmaReduced2.dot(VTReduced2))
print("Matriz A transformada:")
print(AReduced2)
```

```
Matriz A transformada:
[[5.0952927  3.50597743 1.40192232 0.20165319]
 [4.74588049 3.19610853 1.46136967 0.25800276]
 [4.68667405 3.21586325 1.30954904 0.19452725]
 [4.61488457 3.08894388 1.46347879 0.27002699]
 [5.07488651 3.50623125 1.36428119 0.1863997 ]
 [5.52598407 3.7330351  1.67566825 0.28872322]
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 [6.1169056  2.72629392 5.00472583 1.74037685]]
```

[18]:
```
T2 = UReduced2.dot(SigmaReduced2)
print("Matriz T (2 Dimensiones):")
print(T2)
```

```
Matriz T (2 Dimensiones):
[[-5.91274714e+00  2.30203322e+00]
 [-5.57248242e+00  1.97182599e+00]
 [-5.44697714e+00  2.09520636e+00]
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 [-5.51597520e+00  2.07090423e+00]
 [-5.85092859e+00  2.14807482e+00]
 [-5.15891972e+00  1.77506408e+00]
 [-5.64500117e+00  1.99000106e+00]
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 [-5.75200785e+00  2.02037338e+00]
```

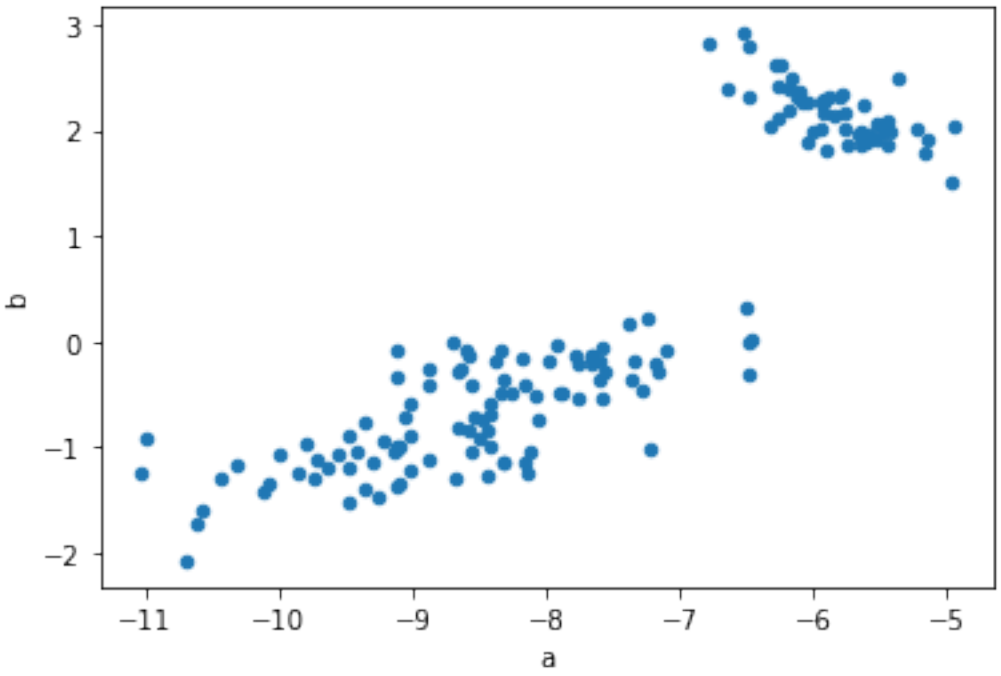
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```

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```

```
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```

[19]:
```
df = pd.DataFrame(T2, columns=["a", "b"])
df.plot(kind = "scatter", x = "a", y = "b")
plt.show
```

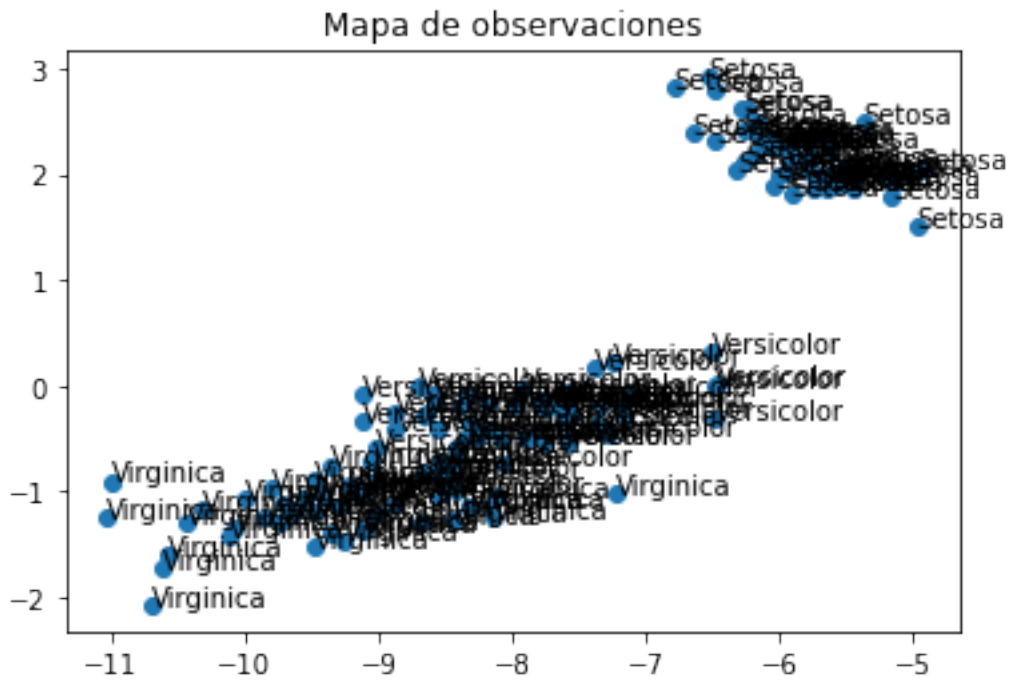
[19]: <function matplotlib.pyplot.show(close=None, block=None)>



[20]:
```
df = pd.DataFrame(T2, columns=["a", "b"])
x = df.iloc[:,0]
y = df.iloc[:, 1]
x = x.to_numpy()
y = y.to_numpy()

fig, ax = plt.subplots()
ax.set_title("Mapa de observaciones")
ax.scatter(x,y)

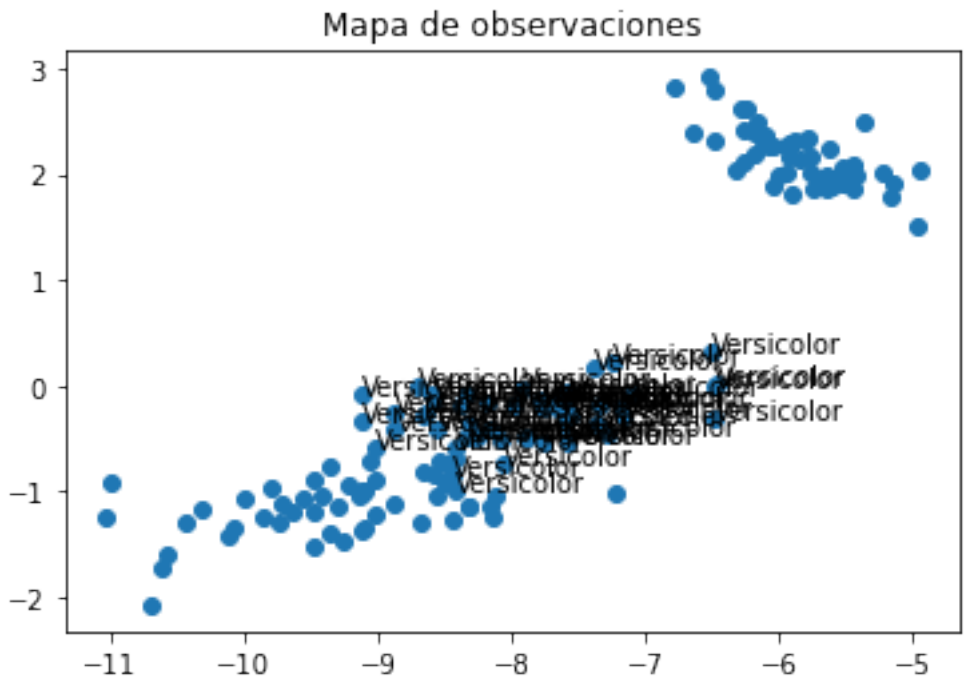
for i, txt in enumerate(etiquetas):
    ax.annotate(txt, (x[i], y[i]))
```



[21]:
```
df = pd.DataFrame(T2, columns=["a", "b"])
x = df.iloc[:,0]
y = df.iloc[:, 1]
x = x.to_numpy()
y = y.to_numpy()

fig, ax = plt.subplots()
ax.set_title("Mapa de observaciones")
ax.scatter(x,y)

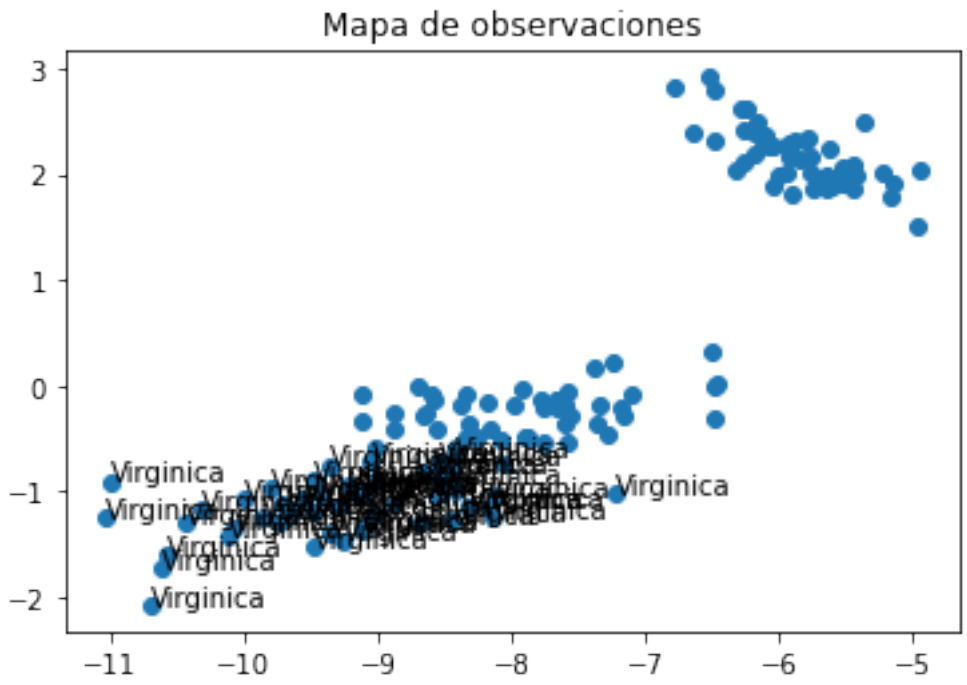
for i, txt in enumerate(etiquetas):
    if txt == "Versicolor":
        ax.annotate(txt, (x[i], y[i]))
```



[22]:
```
df = pd.DataFrame(T2, columns=["a", "b"])
x = df.iloc[:,0]
y = df.iloc[:, 1]
x = x.to_numpy()
y = y.to_numpy()

fig, ax = plt.subplots()
ax.set_title("Mapa de observaciones")
ax.scatter(x,y)

for i, txt in enumerate(etiquetas):
    if txt == "Virginica":
        ax.annotate(txt, (x[i], y[i]))
```



[23]:
```
# Reducción a 3 Dimensiones
n_elements = 3

UReduced3 = U[:, :n_elements]

SigmaReduced3 = Sigma[:n_elements,:n_elements]

VTReduced3 = VT[:n_elements, :]

AReduced3 = UReduced3.dot(SigmaReduced3.dot(VTReduced3))
print("Matriz A transformada:")
print(AReduced3)
```

```
Matriz A transformada:
[[5.09900942 3.50097959 1.4014844  0.19767844]
 [4.86870364 3.03094925 1.44689807 0.12665275]
 [4.69422107 3.20571484 1.30865982 0.18645629]
 [4.62518119 3.0750981  1.4622656  0.25901553]
 [5.01947963 3.58073643 1.37080948 0.24565314]
 [5.40687172 3.89320449 1.68970262 0.41610481]
 [4.61617117 3.38400818 1.37576725 0.33789933]
 [5.01454656 3.38561478 1.47820176 0.23409183]
 [4.40998651 2.89012425 1.38503506 0.22340474]
 [4.91842577 3.08177859 1.4723887  0.14318329]
 [5.40391192 3.69613146 1.49413792 0.20916813]
```

```
[4.8505539  3.3500068  1.5242442  0.31847992]
[4.80311694 2.99691763 1.39532922 0.10730496]
[4.30830887 2.99178329 1.08754903 0.11947296]
[5.74700833 4.05240393 1.27940883 0.07580684]
[5.69603935 4.40391672 1.5059351  0.39071768]
[5.34517978 3.95421221 1.38214894 0.27152138]
[5.07488831 3.52483318 1.43763026 0.24114734]
[5.68993828 3.80995013 1.71507765 0.27641899]
[5.12084537 3.77938583 1.46876289 0.34885395]
[5.40422382 3.39582303 1.69367054 0.20989909]
[5.08654623 3.71330455 1.52016068 0.36846927]
[4.59895639 3.60103203 1.00156386 0.19755416]
[5.05255898 3.34691484 1.77109109 0.38881555]
[4.89682286 3.30425101 1.75490946 0.42691749]
[4.98925744 3.01062341 1.61609789 0.17482335]
[4.98172732 3.41807002 1.6273819  0.35717549]
[5.20414023 3.49590569 1.4937958  0.20970319]
[5.17853921 3.42122276 1.43215932 0.14970374]
[4.74049003 3.15995905 1.53932507 0.29489386]
[4.82001982 3.08020222 1.56999999 0.24691916]
[5.32513562 3.47403405 1.61218543 0.22454518]
[5.2893295  4.01166126 1.36613837 0.30935579]
[5.52908689 4.17123571 1.35641285 0.26816907]
[4.89430466 3.10563218 1.50853456 0.18665219]
[4.94792154 3.25150085 1.27804036 0.07794709]
[5.44241772 3.55694363 1.38628793 0.06504812]
[4.95389292 3.54670481 1.31924064 0.22630535]
[4.40474155 2.99531104 1.29289472 0.21111246]
[5.10425438 3.39579281 1.49362475 0.20997072]
[4.9697575  3.52990708 1.34531885 0.2291226 ]
[4.39908207 2.39979864 1.45122706 0.06348516]
[4.4250976  3.17518076 1.26239086 0.25881963]
[4.94366311 3.55571205 1.68442168 0.46796688]
[5.15841621 3.74223169 1.81246242 0.53690629]
[4.75487471 3.0446248  1.46762094 0.19424276]
[5.16038947 3.74028031 1.50950545 0.34153091]
[4.61993623 3.18028488 1.37012525 0.24672326]
[5.3142041  3.68595344 1.47871493 0.23328924]
[4.98894554 3.31093185 1.41656527 0.17409239]
[6.99242914 3.20748689 4.71134504 1.38225667]
[6.39921513 3.20077617 4.50117614 1.49816055]
[6.89926816 3.10072373 4.90109668 1.49828482]
[5.47136979 2.32831267 4.04290281 1.23290122]
[6.46363383 2.83596283 4.65449526 1.41477095]
[5.7787905  2.72208338 4.38193123 1.48465621]
[6.32641019 3.27388273 4.66042392 1.66189586]
[4.90770332 2.39238212 3.28845646 1.01805377]
[6.61176191 2.88836855 4.5823746  1.32756562]
```

```
[5.20341433 2.69662354 3.89488358 1.40800194]
[4.98754501 2.01231685 3.518664   0.97081004]
[5.88405101 3.0157721  4.22389981 1.46262137]
[5.98209421 2.21770719 4.0268321  0.95803536]
[6.15452467 2.84608007 4.61829395 1.52778595]
[5.56045382 2.93910757 3.65926055 1.20731818]
[6.66685869 3.13277373 4.44966276 1.32232886]
[5.6611965  2.93948223 4.4082961  1.64342229]
[5.86899169 2.63177351 3.9966149  1.16169137]
[6.11801922 2.28107153 4.62284946 1.30786693]
[5.6142529  2.48590518 3.87864181 1.13340361]
[5.92458164 3.175691   4.76316404 1.8576104 ]
[6.06050685 2.83905513 4.05918109 1.20744245]
[6.29995307 2.50004641 4.90007033 1.49989   ]
[6.19258887 2.70843804 4.56125416 1.41699456]
[6.3860773  2.91376828 4.32086338 1.26737027]
[6.56697284 3.03266085 4.4494917  1.32259639]
[6.78772438 2.81213947 4.8183952  1.37123042]
[6.67685524 3.02288805 5.03468277 1.64575713]
[6.00984976 2.89025948 4.48523998 1.52308425]
[5.67656792 2.62317219 3.53511333 0.94508374]
[5.49894407 2.40104422 3.80158233 1.09752528]
[5.5076422  2.39244256 3.68854805 1.01791052]
[5.78990349 2.70998453 3.91512978 1.17633745]
[6.05791052 2.64273178 5.01322021 1.73572114]
[5.48178086 2.91912618 4.37745013 1.69166452]
[6.03661878 3.36378736 4.4451262  1.68582107]
[6.68900654 3.11087153 4.71647387 1.47423534]
[6.25072431 2.34872917 4.47384038 1.18451574]
[5.64774678 2.95278278 4.02845071 1.41190105]
[5.49172584 2.50818239 4.01239895 1.28060839]
[5.58771693 2.51325595 4.26855484 1.40557651]
[6.14927971 2.95126686 4.5261536  1.51549368]
[5.79514845 2.60479774 4.00727013 1.18862973]
[4.98723311 2.31262529 3.31913137 0.97007907]
[5.63263569 2.66772628 4.15109492 1.37648616]
[5.77699871 2.9238553  4.08461626 1.3804569 ]
[5.74269957 2.85777402 4.13601405 1.40007221]
[6.20666166 2.89341223 4.2900174  1.3156125 ]
[5.02690691 2.5722824  3.10953111 0.92869645]
[5.71709855 2.78309109 4.07437756 1.34007277]
[6.30981899 3.29028991 5.98528609 2.52301214]
[5.80613153 2.69393648 5.09081181 1.91437007]
[7.07800896 3.02174714 5.93295392 2.04846101]
[6.37626274 2.8245831  5.48571912 1.97873206]
[6.50021793 2.99978449 5.79967344 2.20051074]
[7.63450897 2.96587377 6.54828778 2.18087645]
[4.93410939 2.46626892 4.44888656 1.77993997]
```

```
[7.38130186 2.81959987 6.17816792 1.99054193]
[6.72522789 2.47505191 5.76219562 1.85912498]
[7.16314645 3.6364448  6.1552256  2.41362872]
[6.4608553  3.23871054 5.15865893 1.90825909]
[6.37522443 2.72450078 5.33712657 1.8419351 ]
[6.74719354 3.05222077 5.57913128 1.97624092]
[5.65652355 2.54299424 5.06515012 1.89810706]
[5.69570398 2.90313926 5.25628917 2.15556816]
[6.32963011 3.26958941 5.40545036 2.13507863]
[6.55043342 2.95012594 5.42442475 1.91819756]
[7.79694288 3.70413232 6.55472961 2.42719877]
[7.68153142 2.61826375 6.92767545 2.25671638]
[6.01571851 2.18445583 4.97644557 1.53683845]
[6.83986116 3.25947183 5.79011897 2.15905649]
[5.58192683 2.81787272 4.92708293 1.95764307]
[7.74340484 2.75707657 6.63495718 2.10172512]
[6.24794578 2.75147688 4.97800405 1.67800388]
[6.71886578 3.28134347 5.67172935 2.1442145 ]
[7.27585916 3.1249822  5.88632389 1.97778622]
[6.152993   2.84648564 4.87044072 1.68983272]
[6.09906421 3.00092541 4.90140229 1.79780686]
[6.38342919 2.81638702 5.62483161 2.06116406]
[7.27289936 2.92790917 5.69075919 1.77084953]
[7.40586457 2.79420048 6.09121185 1.91374441]
[7.97833179 3.722537   6.28261861 2.18358116]
[6.35930808 2.8402406  5.66097747 2.10463297]
[6.36133312 2.73934713 5.00809138 1.64374248]
[6.26279747 2.43900813 5.3560455  1.78153793]
[7.59885963 3.10001861 6.25156039 2.06296384]
[6.28242619 3.4173789  5.62633463 2.35881339]
[6.47090363 3.02988277 5.39374983 1.96617225]
[5.99393341 3.00599931 4.80909088 1.78578212]
[6.8316564  3.16758559 5.50241392 1.93982753]
[6.61072339 3.18828643 5.73378236 2.19076818]
[6.73714521 3.26104854 5.34404039 1.91832775]
[5.80613153 2.69393648 5.09081181 1.91437007]
[6.78099932 3.21878995 5.92847282 2.25546932]
[6.62238132 3.3767578  5.81631278 2.31809011]
[6.56297453 3.1355057  5.40533476 1.97886226]
[6.2188916  2.58020882 5.12154218 1.70991147]
[6.45592223 3.0435889  5.26605121 1.89669778]
[6.18599351 3.41385115 5.42098894 2.26717389]
[5.95049455 2.9500655  5.02433315 1.91834081]]
```

[24]:
```
T3 = UReduced3.dot(SigmaReduced3)
print("Matriz T (3 Dimensiones):")
print(T3)
```

```
Matriz T (3 Dimensiones):
[[-5.91274714e+00  2.30203322e+00  7.40153559e-03]
 [-5.57248242e+00  1.97182599e+00  2.44592251e-01]
 [-5.44697714e+00  2.09520636e+00  1.50292625e-02]
 [-5.43645948e+00  1.87038151e+00  2.05048805e-02]
 [-5.87564494e+00  2.32829018e+00 -1.10338269e-01]
 [-6.47759822e+00  2.32464996e+00 -2.37202487e-01]
 [-5.51597520e+00  2.07090423e+00 -2.29853120e-01]
 [-5.85092859e+00  2.14807482e+00  1.87937744e-02]
 [-5.15891972e+00  1.77506408e+00  6.10392203e-02]
 [-5.64500117e+00  1.99000106e+00  2.24852923e-01]
 [-6.26539771e+00  2.42576813e+00  1.70826682e-02]
 [-5.75200785e+00  2.02037338e+00 -8.75537913e-02]
 [-5.48058085e+00  1.97777558e+00  2.48078403e-01]
 [-4.95112411e+00  2.04828749e+00  1.47509030e-02]
 [-6.52596417e+00  2.91606081e+00  3.31214244e-02]
 [-6.79037199e+00  2.82500759e+00 -4.12344488e-01]
 [-6.27239468e+00  2.60811578e+00 -2.13536004e-01]
 [-5.92953789e+00  2.26766614e+00 -4.63000893e-02]
 [-6.64813130e+00  2.38959506e+00  3.46698872e-02]
 [-6.09486463e+00  2.36082303e+00 -2.54789706e-01]
 [-6.25397363e+00  2.12001187e+00  2.07822423e-01]
 [-6.07364677e+00  2.27178150e+00 -2.40966999e-01]
 [-5.36999813e+00  2.49808604e+00 -2.87533676e-01]
 [-6.04100482e+00  1.87698371e+00 -3.64045375e-02]
 [-5.90591050e+00  1.80777402e+00 -1.05303653e-01]
 [-5.75019501e+00  1.85851056e+00  2.82974481e-01]
 [-5.93581099e+00  2.00847420e+00 -9.45260962e-02]
 [-6.03915884e+00  2.25958425e+00  5.17003873e-02]
 [-5.94984934e+00  2.27577626e+00  1.25141340e-01]
 [-5.60087980e+00  1.88260699e+00 -2.72059978e-03]
 [-5.63798200e+00  1.85635003e+00  1.15019205e-01]
 [-6.18495337e+00  2.19301062e+00  1.12252414e-01]
 [-6.25041979e+00  2.62199803e+00 -2.99743980e-01]
 [-6.47925073e+00  2.79842433e+00 -2.64406898e-01]
 [-5.66179193e+00  1.95563398e+00  1.71151298e-01]
 [-5.62100870e+00  2.25132529e+00  1.71592301e-01]
 [-6.16188952e+00  2.48656963e+00  2.14180046e-01]
 [-5.78374337e+00  2.33423977e+00 -1.06852116e-01]
 [-5.14562746e+00  1.90060499e+00 -5.68490950e-04]
 [-5.92603941e+00  2.17649231e+00  6.90092468e-02]
 [-5.80312619e+00  2.31011510e+00 -9.05989410e-02]
 [-4.97146870e+00  1.51193425e+00  4.68615680e-01]
 [-5.22164469e+00  2.00995389e+00 -1.35617155e-01]
 [-6.00740111e+00  1.99441449e+00 -2.69453678e-01]
 [-6.31685893e+00  2.04299013e+00 -3.32157814e-01]
 [-5.51416236e+00  1.90904141e+00  1.40675153e-01]
 [-6.12937476e+00  2.32432366e+00 -2.07004702e-01]
```

```
[-5.42316721e+00  1.99592242e+00 -4.11028307e-02]
[-6.19028689e+00  2.39735064e+00 -3.31328042e-02]
[-5.76161909e+00  2.16426683e+00  9.22347271e-02]
[-9.12024508e+00 -7.30558200e-02  3.24400524e-01]
[-8.58376916e+00 -1.36194931e-01 -1.87606940e-02]
[-9.12651817e+00 -3.32247752e-01  2.76174517e-01]
[-7.27560832e+00 -4.60955955e-01  2.74005395e-01]
[-8.55814639e+00 -3.97341697e-01  2.95635486e-01]
[-7.87237747e+00 -4.85081002e-01  7.23157664e-03]
[-8.66605949e+00 -2.86037963e-01 -2.02035365e-01]
[-6.45347358e+00  2.23799845e-02  1.07709449e-01]
[-8.63768432e+00 -2.45515595e-01  3.85729876e-01]
[-7.16780021e+00 -2.91011251e-01 -1.94523354e-01]
[-6.47915170e+00 -3.09633237e-01  4.16189007e-01]
[-7.97829519e+00 -1.75031916e-01 -1.17039530e-01]
[-7.56278153e+00 -2.70441712e-01  7.53711964e-01]
[-8.33022188e+00 -4.92836582e-01  7.50342680e-02]
[-7.37356730e+00  1.78974058e-01 -5.72586411e-02]
[-8.70300135e+00 -3.83374315e-04  2.59028301e-01]
[-7.90686540e+00 -4.72883753e-01 -2.85435810e-01]
[-7.65390386e+00 -1.24770897e-01  3.09742739e-01]
[-8.05346136e+00 -7.39774413e-01  5.56051681e-01]
[-7.34185398e+00 -1.83588948e-01  3.02492074e-01]
[-8.41249000e+00 -5.93982991e-01 -4.48692793e-01]
[-7.91631631e+00 -1.70787632e-02  2.37676570e-01]
[-8.44780157e+00 -8.30799394e-01  3.80027674e-01]
[-8.25863175e+00 -4.78776871e-01  2.49961850e-01]
[-8.33356003e+00 -8.97512089e-02  3.03048793e-01]
[-8.58988192e+00 -8.34753146e-02  2.76337160e-01]
[-8.86928986e+00 -4.19455056e-01  4.88150286e-01]
[-9.02317031e+00 -5.83357800e-01  1.29948034e-01]
[-8.16930004e+00 -4.13888242e-01 -1.70495879e-02]
[-7.23297912e+00  2.17335895e-01  3.62551323e-01]
[-7.17743366e+00 -1.95814432e-01  3.25717555e-01]
[-7.10934202e+00 -9.05808962e-02  3.85335800e-01]
[-7.58488360e+00 -5.17721477e-02  2.14172730e-01]
[-8.41787888e+00 -9.82802956e-01  2.87977267e-02]
[-7.75664376e+00 -5.29718734e-01 -3.85866754e-01]
[-8.37613388e+00 -1.74883073e-01 -4.08372873e-01]
[-8.87369476e+00 -2.47349822e-01  1.87576814e-01]
[-8.08169840e+00 -5.17081855e-01  6.52062692e-01]
[-7.66808034e+00 -1.20683770e-01 -1.54366077e-01]
[-7.35162556e+00 -3.51607055e-01  1.38956731e-01]
[-7.57804696e+00 -5.46031346e-01  1.01467541e-01]
[-8.31692961e+00 -3.67295677e-01  1.34265568e-02]
[-7.59817587e+00 -1.77313053e-01  2.75780442e-01]
[-6.49057578e+00 -3.87697543e-03  2.25449253e-01]
[-7.60535538e+00 -3.55573575e-01  4.22902984e-02]
```

[-7.77770129e+00 -1.28765654e-01 -5.63656001e-02]
[-7.75648343e+00 -2.17807185e-01 -4.25428931e-02]
[-8.18333840e+00 -1.46586189e-01 2.02617848e-01]
[-6.50459193e+00 3.12121701e-01 1.04664299e-01]
[-7.66717393e+00 -2.01615180e-01 3.08980596e-02]
[-9.48408779e+00 -1.51660561e+00 -7.62266059e-01]
[-8.31802951e+00 -1.14273918e+00 -2.32738093e-01]
[-9.85248456e+00 -1.24495426e+00 6.27538370e-02]
[-9.00931450e+00 -1.21126802e+00 -9.25908735e-02]
[-9.36730953e+00 -1.37895983e+00 -2.86324002e-01]
[-1.05871448e+01 -1.59893200e+00 2.72414854e-01]
[-7.22462810e+00 -1.01391260e+00 -4.06725707e-01]
[-1.01195289e+01 -1.42315831e+00 3.68147506e-01]
[-9.26032507e+00 -1.45802877e+00 3.66535103e-01]
[-1.03254119e+01 -1.16769130e+00 -5.18816424e-01]
[-9.05063906e+00 -7.04811578e-01 -2.72553071e-01]
[-8.87129618e+00 -1.11396715e+00 5.67215001e-02]
[-9.42194857e+00 -1.04674091e+00 -6.42260973e-02]
[-8.13239132e+00 -1.24400619e+00 -1.95689906e-01]
[-8.43999189e+00 -1.25990013e+00 -5.68770550e-01]
[-9.12850228e+00 -9.78063221e-01 -4.95706659e-01]
[-9.14624386e+00 -1.02889214e+00 -5.37676399e-02]
[-1.10344162e+01 -1.23835244e+00 -2.77182575e-01]
[-1.06977054e+01 -2.07054584e+00 4.67574542e-01]
[-8.15974415e+00 -1.15094167e+00 4.26037632e-01]
[-9.70925990e+00 -1.11944159e+00 -2.68295780e-01]
[-8.12000547e+00 -1.03753388e+00 -4.42561753e-01]
[-1.06207486e+01 -1.71636278e+00 5.05463995e-01]
[-8.57419106e+00 -8.24551736e-01 8.38741356e-02]
[-9.56346538e+00 -1.05286796e+00 -3.28847807e-01]
[-1.00045412e+01 -1.07495308e+00 1.33108900e-01]
[-8.48578798e+00 -7.27428321e-01 -2.79490480e-02]
[-8.53799528e+00 -7.17363366e-01 -2.19129805e-01]
[-9.09678896e+00 -1.34062623e+00 -1.35955944e-01]
[-9.79234073e+00 -9.73834912e-01 3.87394055e-01]
[-1.00708200e+01 -1.34204944e+00 4.44018927e-01]
[-1.09971537e+01 -9.00183935e-01 -5.15985179e-02]
[-9.11357971e+00 -1.37499331e+00 -1.89657569e-01]
[-8.66442919e+00 -8.08508955e-01 1.65621437e-01]
[-8.67790400e+00 -1.29465803e+00 2.24357677e-01]
[-1.04393327e+01 -1.28491639e+00 2.44810180e-01]
[-9.30010211e+00 -1.14409826e+00 -7.52422283e-01]
[-9.10914166e+00 -1.00263518e+00 -1.71507444e-01]
[-8.41158358e+00 -6.74914401e-01 -2.63428657e-01]
[-9.48376712e+00 -8.92782514e-01 -7.56183361e-02]
[-9.48651952e+00 -1.19445165e+00 -3.48987397e-01]
[-9.36344597e+00 -7.48917309e-01 -1.65271724e-01]
[-8.31802951e+00 -1.14273918e+00 -2.32738093e-01]

```
[-9.73675086e+00 -1.28959199e+00 -3.30344494e-01]
[-9.63062839e+00 -1.19033628e+00 -5.43654307e-01]
[-9.22651660e+00 -9.31293195e-01 -2.04094957e-01]
[-8.56626547e+00 -1.03913417e+00  1.59304554e-01]
[-9.02592271e+00 -8.85026933e-01 -1.43421027e-01]
[-9.10559876e+00 -9.96415757e-01 -7.37102889e-01]
[-8.49037542e+00 -9.15931258e-01 -3.31393992e-01]]
```

[25]:
```
from mpl_toolkits import mplot3d
%matplotlib inline
```

[26]:
```
df3 = pd.DataFrame(T3, columns = ["a", "b", "c"])
df3
```

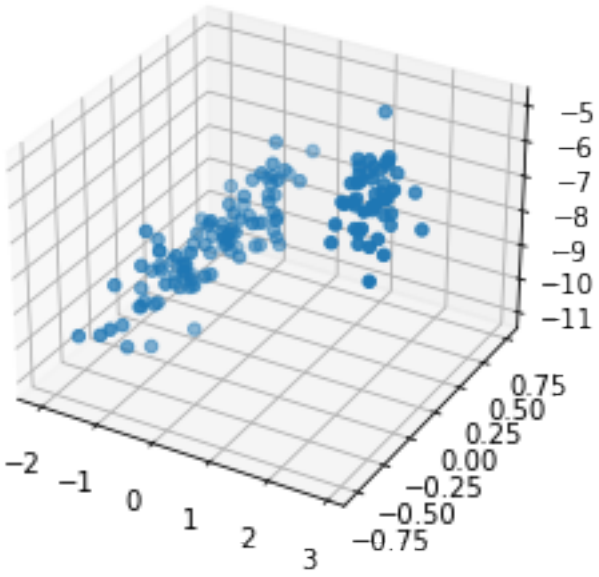
[26]:

	a	b	c
0	-5.912747	2.302033	0.007402
1	-5.572482	1.971826	0.244592
2	-5.446977	2.095206	0.015029
3	-5.436459	1.870382	0.020505
4	-5.875645	2.328290	-0.110338
..	…	…	…
145	-9.226517	-0.931293	-0.204095
146	-8.566265	-1.039134	0.159305
147	-9.025923	-0.885027	-0.143421
148	-9.105599	-0.996416	-0.737103
149	-8.490375	-0.915931	-0.331394

[150 rows x 3 columns]

[27]:
```
fig = plt.figure()
ax = plt.axes(projection="3d")
xline = df3["b"]
yline = df3["c"]
zline = df3["a"]
ax.scatter3D(xline, yline, zline)
```

[27]:
```
<mpl_toolkits.mplot3d.art3d.Path3DCollection at 0x723956856ee0>
```



[28]:
```
from mpl_toolkits.mplot3d import Axes3D

ax3d = plt.figure().gca(projection="3d")

arrayx = np.array(df3["a"])
arrayy = np.array(df3["b"])
arrayz = np.array(df3["c"])

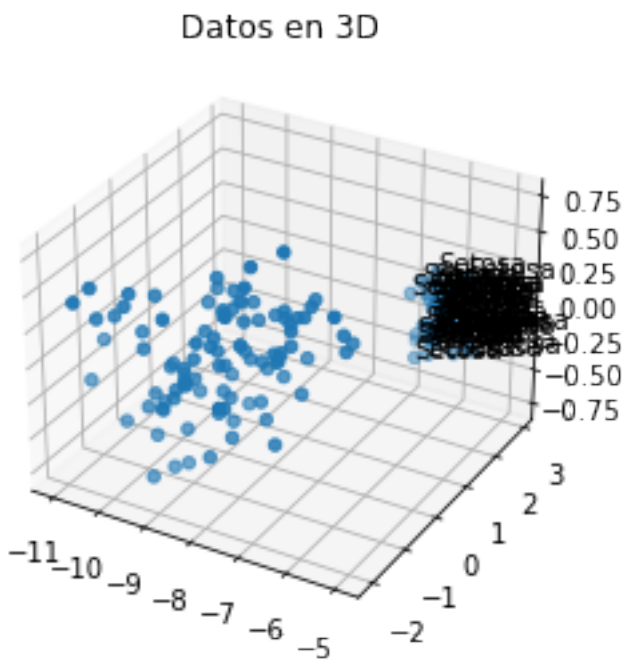
ax3d.scatter(arrayx, arrayy, arrayz)

for x, y, z, label in zip(arrayx, arrayy, arrayz, etiquetas):
    if label == "Setosa":
        ax3d.text(x,y,z,label)

plt.title("Datos en 3D")
plt.show
```

```
/tmp/ipykernel_849/879455700.py:3: MatplotlibDeprecationWarning: Calling gca()
with keyword arguments was deprecated in Matplotlib 3.4. Starting two minor
releases later, gca() will take no keyword arguments. The gca() function should
only be used to get the current axes, or if no axes exist, create new axes with
default keyword arguments. To create a new axes with non-default arguments, use
plt.axes() or plt.subplot().
  ax3d = plt.figure().gca(projection="3d")
```

[28]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```



[29]:
```
ax3d = plt.figure().gca(projection="3d")

arrayx = np.array(df3["a"])
arrayy = np.array(df3["b"])
arrayz = np.array(df3["c"])

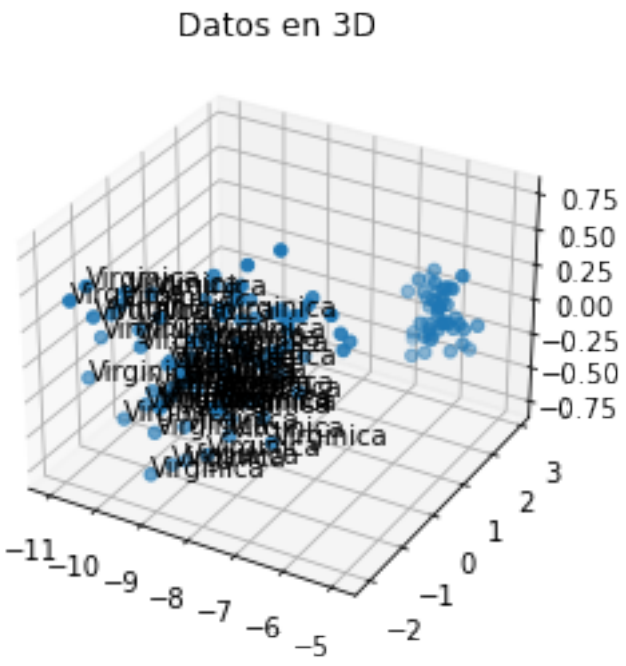
ax3d.scatter(arrayx, arrayy, arrayz)

for x, y, z, label in zip(arrayx, arrayy, arrayz, etiquetas):
    if label == "Virginica":
        ax3d.text(x,y,z,label)

plt.title("Datos en 3D")
plt.show
```

```
/tmp/ipykernel_849/3258452029.py:1: MatplotlibDeprecationWarning: Calling gca()
with keyword arguments was deprecated in Matplotlib 3.4. Starting two minor
releases later, gca() will take no keyword arguments. The gca() function should
only be used to get the current axes, or if no axes exist, create new axes with
default keyword arguments. To create a new axes with non-default arguments, use
plt.axes() or plt.subplot().
  ax3d = plt.figure().gca(projection="3d")
```

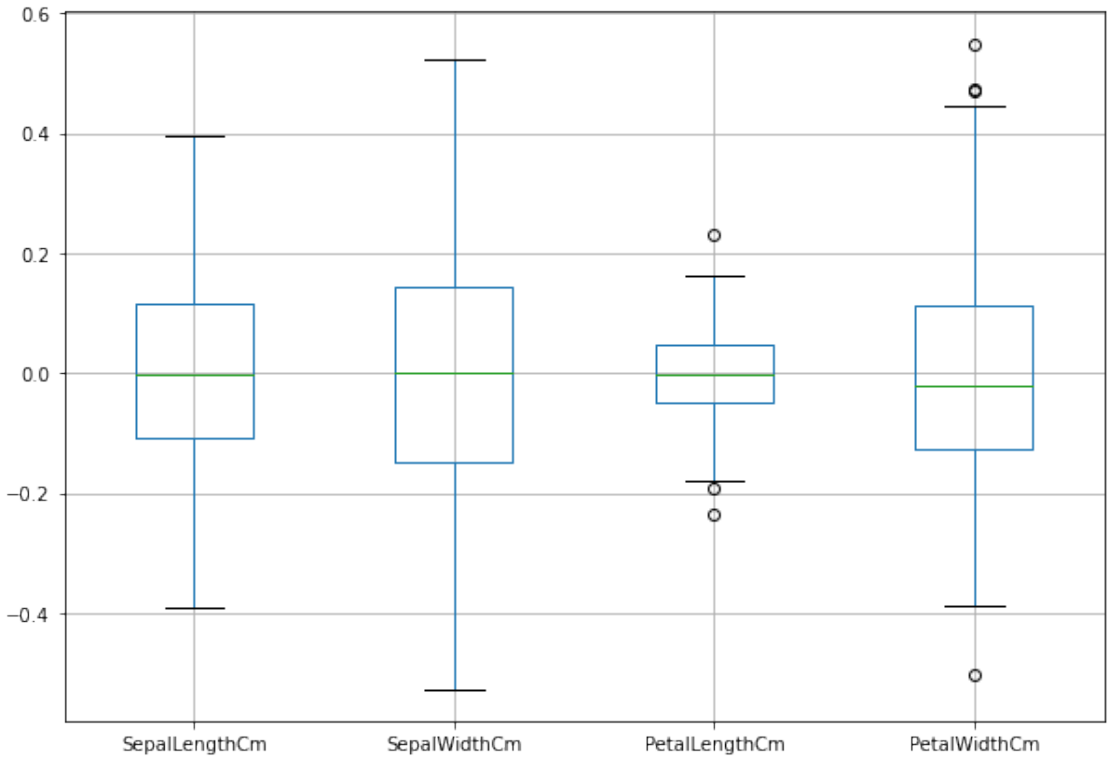
[29]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```



[31]:
```
plt.figure(figsize = (10,7))
Error2 = B - AReduced2

dfError2 = pd.DataFrame(Error2, columns = ["SepalLengthCm", "SepalWidthCm",␣
↪"PetalLengthCm", "PetalWidthCm"])
dfError2.boxplot()
```

[31]: <AxesSubplot:>



[32]:
```
print("Errores promedio por columna")
Error2.mean(axis=0)
```

```
Errores promedio por columna
```

[32]:
```
array([ 1.63686453e-03, -2.22711818e-03, -9.18095159e-05, -2.00030563e-03])
```

[33]:
```
Error3 = B - AReduced3
print(Error3)
```

```
[[ 9.90580176e-04 -9.79593534e-04 -1.48439954e-03  2.32155879e-03]
 [ 3.12963572e-02 -3.09492456e-02 -4.68980698e-02  7.33472515e-02]
 [ 5.77893329e-03 -5.71483845e-03 -8.65981989e-03  1.35437128e-02]
 [-2.51811925e-02  2.49019050e-02  3.77344021e-02 -5.90155350e-02]
 [-1.94796253e-02  1.92635745e-02  2.91905164e-02 -4.56531399e-02]
 [-6.87172359e-03  6.79550850e-03  1.02973829e-02 -1.61048149e-02]
 [-1.61711716e-02  1.59918152e-02  2.42327479e-02 -3.78993307e-02]
 [-1.45465586e-02  1.43852210e-02  2.17982405e-02 -3.40918301e-02]
 [-9.98650905e-03  9.87574750e-03  1.49649365e-02 -2.34047364e-02]
 [-1.84257711e-02  1.82214087e-02  2.76112998e-02 -4.31832900e-02]
 [-3.91192470e-03  3.86853708e-03  5.86207898e-03 -9.16812532e-03]
 [-5.05539029e-02  4.99932036e-02  7.57557965e-02 -1.18479918e-01]
 [-3.11693586e-03  3.08236556e-03  4.67077605e-03 -7.30496132e-03]
```

```
[-8.30886785e-03  8.21671322e-03  1.24509655e-02 -1.94729571e-02]
[ 5.29916703e-02 -5.24039334e-02 -7.94088282e-02  1.24193156e-01]
[ 3.96065208e-03 -3.91672402e-03 -5.93509770e-03  9.28232452e-03]
[ 5.48202245e-02 -5.42122069e-02 -8.21489447e-02  1.28478620e-01]
[ 2.51116946e-02 -2.48331778e-02 -3.76302583e-02  5.88526571e-02]
[ 1.00617247e-02 -9.95012889e-03 -1.50776483e-02  2.35810144e-02]
[-2.08453730e-02  2.06141745e-02  3.12371101e-02 -4.88539546e-02]
[-4.22381822e-03  4.17697135e-03  6.32945619e-03 -9.89909002e-03]
[ 1.34537683e-02 -1.33045510e-02 -2.01606775e-02  3.15307281e-02]
[ 1.04360837e-03 -1.03203358e-03 -1.56386310e-03  2.44583755e-03]
[ 4.74410161e-02 -4.69148422e-02 -7.10910880e-02  1.11184446e-01]
[-9.68228640e-02  9.57489902e-02  1.45090542e-01 -2.26917494e-01]
[ 1.07425617e-02 -1.06234147e-02 -1.60978930e-02  2.51766483e-02]
[ 1.82726832e-02 -1.80700187e-02 -2.73818952e-02  4.28245077e-02]
[-4.14022824e-03  4.09430848e-03  6.20419532e-03 -9.70318558e-03]
[ 2.14607856e-02 -2.12227615e-02 -3.21593154e-02  5.02962575e-02]
[-4.04900278e-02  4.00409481e-02  6.06749258e-02 -9.48938636e-02]
[-2.00198223e-02  1.97977801e-02  3.00000099e-02 -4.69191649e-02]
[ 7.48643847e-02 -7.40340550e-02 -1.12185425e-01  1.75454824e-01]
[-8.93295038e-02  8.83387397e-02  1.33861627e-01 -2.09355790e-01]
[-2.90868933e-02  2.87642871e-02  4.35871544e-02 -6.81690736e-02]
[ 5.69534331e-03 -5.63217558e-03 -8.53455903e-03  1.33478083e-02]
[ 5.20784562e-02 -5.15008479e-02 -7.80403629e-02  1.22052915e-01]
[ 5.75822817e-02 -5.69436298e-02 -8.62879296e-02  1.34951875e-01]
[-5.38929183e-02  5.32951856e-02  8.07593622e-02 -1.26305353e-01]
[-4.74154887e-03  4.68895979e-03  7.10528348e-03 -1.11124619e-02]
[-4.25438001e-03  4.20719418e-03  6.37525349e-03 -9.97071571e-03]
[ 3.02425030e-02 -2.99070799e-02 -4.53188532e-02  7.08774015e-02]
[ 1.00917932e-01 -9.97986394e-02 -1.51227064e-01  2.36514841e-01]
[-2.50976026e-02  2.48192421e-02  3.76091412e-02 -5.88196305e-02]
[ 5.63368852e-02 -5.57120462e-02 -8.44216839e-02  1.32033120e-01]
[-5.84162067e-02  5.77683056e-02  8.75375789e-02 -1.36906291e-01]
[ 4.51252930e-02 -4.46248030e-02 -6.76209416e-02  1.05757235e-01]
[-6.03894744e-02  5.97196876e-02  9.04945508e-02 -1.41530912e-01]
[-1.99362324e-02  1.97151173e-02  2.98747490e-02 -4.67232605e-02]
[-1.42041033e-02  1.40465639e-02  2.12850660e-02 -3.32892398e-02]
[ 1.10544553e-02 -1.09318490e-02 -1.65652702e-02  2.59076130e-02]
[ 7.57085563e-03 -7.48688637e-03 -1.13450429e-02  1.77433255e-02]
[ 7.84872425e-04 -7.76167312e-04 -1.17614333e-03  1.83945482e-03]
[ 7.31844232e-04 -7.23727261e-04 -1.09667977e-03  1.71517607e-03]
[ 2.86302128e-02 -2.83126717e-02 -4.29028117e-02  6.70987810e-02]
[ 3.63661714e-02 -3.59628299e-02 -5.44952639e-02  8.52290477e-02]
[-7.87904994e-02  7.79166246e-02  1.18068768e-01 -1.84656206e-01]
[-2.64101927e-02  2.61172741e-02  3.95760775e-02 -6.18958632e-02]
[-7.70331983e-03  7.61788140e-03  1.15435426e-02 -1.80537733e-02]
[-1.17619057e-02  1.16314530e-02  1.76253955e-02 -2.75656189e-02]
[-3.41432898e-03  3.37646025e-03  5.11642418e-03 -8.00194236e-03]
[ 1.24549921e-02 -1.23168523e-02 -1.86639961e-02  2.91899608e-02]
```

```
[ 1.59489941e-02 -1.57721020e-02 -2.38998116e-02  3.73786277e-02]
[ 1.79057894e-02 -1.77071942e-02 -2.68320992e-02  4.19646425e-02]
[-5.45246714e-02  5.39199318e-02  8.17060538e-02 -1.27785952e-01]
[ 3.95461784e-02 -3.91075671e-02 -5.92605531e-02  9.26818248e-02]
[ 3.31413077e-02 -3.27737336e-02 -4.96627564e-02  7.76711429e-02]
[-6.11965028e-02  6.05177652e-02  9.17038952e-02 -1.43422292e-01]
[-6.89916891e-02  6.82264941e-02  1.03385101e-01 -1.61691367e-01]
[ 8.19807837e-02 -8.10715251e-02 -1.22849458e-01  1.92133069e-01]
[-1.42529041e-02  1.40948235e-02  2.13581948e-02 -3.34036110e-02]
[-2.45816384e-02  2.43090006e-02  3.68359611e-02 -5.76103987e-02]
[ 3.94931503e-02 -3.90551271e-02 -5.91810895e-02  9.25575461e-02]
[ 4.69336247e-05 -4.64130783e-05 -7.03307542e-05  1.09995306e-04]
[-9.25888734e-02  9.15619593e-02  1.38745843e-01 -2.16994564e-01]
[ 1.39226982e-02 -1.37682799e-02 -2.08633761e-02  3.26297287e-02]
[ 3.30271559e-02 -3.26608479e-02 -4.94916982e-02  7.74036128e-02]
[ 1.22756188e-02 -1.21394684e-02 -1.83952024e-02  2.87695750e-02]
[ 2.31447557e-02 -2.28880545e-02 -3.46827703e-02  5.42428694e-02]
[-9.84976149e-03  9.74051663e-03  1.47600182e-02 -2.30842500e-02]
[ 2.34320813e-02 -2.31721934e-02 -3.51133322e-02  5.49162560e-02]
[ 1.05593114e-03 -1.04421969e-03 -1.58232896e-03  2.47471764e-03]
[-7.64219626e-03  7.55743576e-03  1.14519479e-02 -1.79105219e-02]
[ 1.00965139e-02 -9.98453222e-03 -1.51297804e-02  2.36625476e-02]
[-5.79105155e-02  5.72682232e-02  8.67797930e-02 -1.35721136e-01]
[-8.17808600e-02  8.08738189e-02  1.22549869e-01 -1.91664521e-01]
[-3.66187813e-02  3.62126381e-02  5.48738037e-02 -8.58210732e-02]
[ 1.09934611e-02 -1.08715313e-02 -1.64738695e-02  2.57646648e-02]
[ 4.92756936e-02 -4.87291711e-02 -7.38403803e-02  1.15484261e-01]
[-4.77467836e-02  4.72172183e-02  7.15492852e-02 -1.11901053e-01]
[ 8.27415910e-03 -8.18238943e-03 -1.23989539e-02  1.93916124e-02]
[-8.77169303e-02  8.67440515e-02  1.31445161e-01 -2.05576505e-01]
[-4.92797112e-02  4.87331441e-02  7.38464007e-02 -1.15493677e-01]
[ 4.85155367e-03 -4.79774451e-03 -7.27012736e-03  1.13702731e-02]
[ 1.27668856e-02 -1.26252866e-02 -1.91313733e-02  2.99209255e-02]
[-3.26356900e-02  3.22737237e-02  4.89050805e-02 -7.64861594e-02]
[-7.69987064e-02  7.61447047e-02  1.15383739e-01 -1.80456896e-01]
[-4.26995651e-02  4.22259792e-02  6.39859512e-02 -1.00072214e-01]
[-6.66165907e-03  6.58777382e-03  9.98259798e-03 -1.56125002e-02]
[ 7.30930861e-02 -7.22824020e-02 -1.09531107e-01  1.71303546e-01]
[-1.70985512e-02  1.69089092e-02  2.56224405e-02 -4.00727705e-02]
[-9.81899416e-03  9.71009054e-03  1.47139129e-02 -2.30121426e-02]
[-6.13152948e-03  6.06352396e-03  9.18819068e-03 -1.43700697e-02]
[ 2.19910446e-02 -2.17471393e-02 -3.29539166e-02  5.15389912e-02]
[-7.62627397e-02  7.54169006e-02  1.14280882e-01 -1.78732059e-01]
[-2.17925623e-04  2.15508584e-04  3.26564878e-04 -5.10738210e-04]
[-3.45089715e-02  3.41262284e-02  5.17122215e-02 -8.08764482e-02]
[-3.41093900e-02  3.37310787e-02  5.11134425e-02 -7.99399748e-02]
[-8.13018627e-02  8.04001342e-02  1.21832084e-01 -1.90541926e-01]
[-2.52278919e-02  2.49480864e-02  3.78043817e-02 -5.91249812e-02]
```

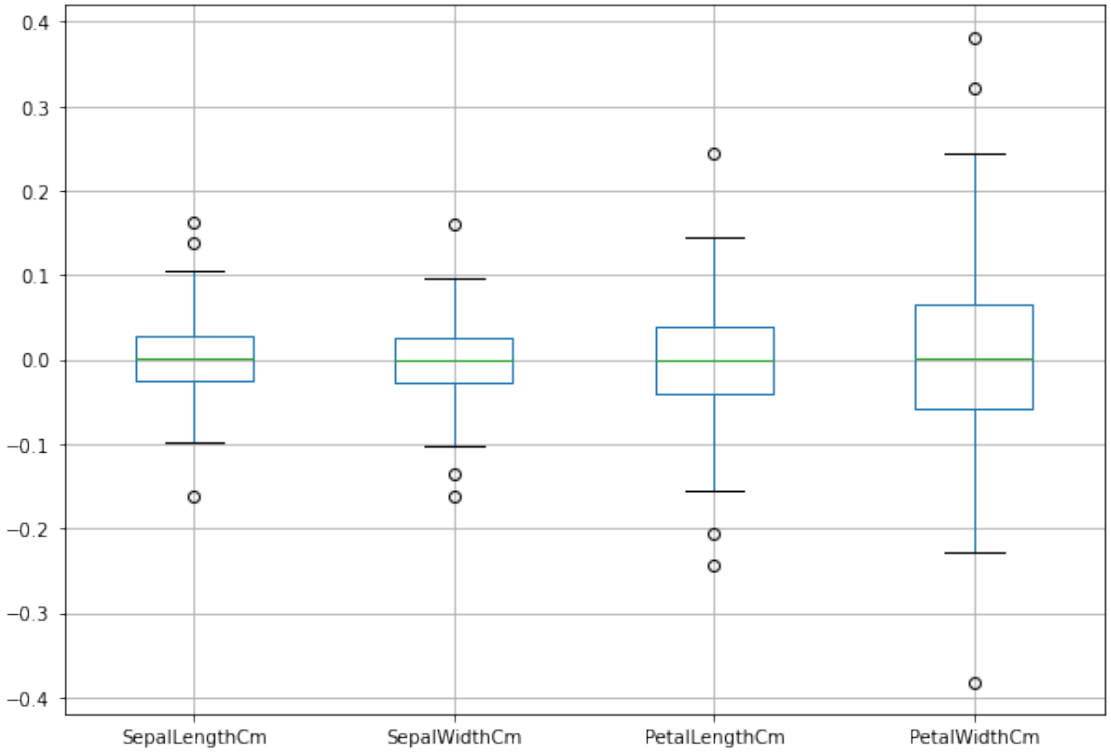
```
[ 3.68535458e-02 -3.64447988e-02 -5.52256019e-02  8.63712756e-02]
[ 3.91447010e-02 -3.87105425e-02 -5.86589330e-02  9.17409081e-02]
[ 2.47755682e-02 -2.45007794e-02 -3.71265677e-02  5.80648993e-02]
[ 5.28064569e-02 -5.22207742e-02 -7.91312831e-02  1.23759083e-01]
[ 4.34764471e-02 -4.29942447e-02 -6.51501207e-02  1.01892942e-01]
[ 1.04296016e-01 -1.03139256e-01 -1.56289174e-01  2.44431838e-01]
[ 7.03698916e-02 -6.95894109e-02 -1.05450359e-01  1.64921371e-01]
[-5.04334223e-02  4.98740592e-02  7.55752544e-02 -1.18197555e-01]
[-9.69428803e-02  9.58676754e-02  1.45270388e-01 -2.27198769e-01]
[ 1.84685823e-02 -1.82637450e-02 -2.76754529e-02  4.32836237e-02]
[-1.57185087e-02  1.55441729e-02  2.35544256e-02 -3.68384539e-02]
[ 6.01388366e-02 -5.94718297e-02 -9.01189662e-02  1.40943508e-01]
[ 1.80731749e-02 -1.78727232e-02 -2.70829290e-02  4.23569331e-02]
[-4.34048407e-02  4.29234325e-02  6.50428174e-02 -1.01725122e-01]
[ 5.20542232e-02 -5.14768837e-02 -7.80040494e-02  1.21996122e-01]
[-1.88657763e-02  1.86565338e-02  2.82706543e-02 -4.42145018e-02]
[-7.58591608e-02  7.50177979e-02  1.13676112e-01 -1.77786217e-01]
[ 4.70070048e-02 -4.64856446e-02 -7.04407155e-02  1.10167282e-01]
[ 9.35785452e-04 -9.25406545e-04 -1.40228881e-03  2.19313994e-03]
[ 1.65708091e-02 -1.63870203e-02 -2.48316108e-02  3.88359352e-02]
[-7.28993619e-02  7.20908265e-02  1.09240808e-01 -1.70849527e-01]
[-5.86456877e-03  5.79952415e-03  8.78814598e-03 -1.37444111e-02]
[-7.83317908e-02  7.74630037e-02  1.17381386e-01 -1.83581160e-01]
[ 4.06919235e-02 -4.02406046e-02 -6.09774696e-02  9.53670336e-02]
[-6.13331210e-02  6.06528681e-02  9.19086196e-02 -1.43742475e-01]
[-1.62797474e-01  1.60991868e-01  2.43954504e-01 -3.81537928e-01]
[ 1.01140371e-01 -1.00018611e-01 -1.51560393e-01  2.37036157e-01]
[ 1.75738127e-02 -1.73788994e-02 -2.63346270e-02  4.11866100e-02]
[-7.09036277e-02  7.01172272e-02  1.06250170e-01 -1.66172254e-01]
[ 6.06659387e-03 -5.99930856e-03 -9.09088367e-03  1.42178843e-02]
[ 6.83435957e-02 -6.75855888e-02 -1.02413923e-01  1.60172472e-01]
[ 8.92766077e-02 -8.82864303e-02 -1.33782362e-01  2.09231821e-01]
[ 1.62854786e-01 -1.61048544e-01 -2.44040386e-01  3.81672245e-01]
[-6.13152948e-03  6.06352396e-03  9.18819068e-03 -1.43700697e-02]
[ 1.90006839e-02 -1.87899451e-02 -2.84728154e-02  4.45306760e-02]
[ 7.76186814e-02 -7.67578034e-02 -1.16312781e-01  1.81909892e-01]
[ 1.37025468e-01 -1.35505703e-01 -2.05334759e-01  3.21137742e-01]
[ 8.11084043e-02 -8.02088214e-02 -1.21542184e-01  1.90088530e-01]
[ 4.40777677e-02 -4.35888960e-02 -6.60512089e-02  1.03302218e-01]
[ 1.40064937e-02 -1.38511460e-02 -2.09889450e-02  3.28261148e-02]
[-5.04945458e-02  4.99345049e-02  7.56668490e-02 -1.18340807e-01]]
```

[34]:
```
plt.figure(figsize = (10,7))


dfError3 = pd.DataFrame(Error3, columns = ["SepalLengthCm", "SepalWidthCm",␣
 ↪"PetalLengthCm", "PetalWidthCm"])
```

```
dfError3.boxplot()
```

[34]: <AxesSubplot:>



[35]:
```
print("Errores promedio por columna")
Error3.mean(axis=0)
```

```
Errores promedio por columna
```

[35]: array([-7.31905331e-05, 7.23787682e-05, 1.09677133e-04, -1.71531926e-04])

[36]:
```
#Comparar las reducciones
print("Error de reconstrucción con 2 dimensiones:")
print(np.linalg.norm(Error2))

print("Error de reconstrucción con 3 dimensiones:")
print(np.linalg.norm(Error3))
```

```
Error de reconstrucción con 2 dimensiones:
3.9408898878793757
Error de reconstrucción con 3 dimensiones:
1.884826305918045
```

[ ]: